

Total No. of Questions :6]

SEAT No. :

P199

[Total No. of Pages :2

Oct./ BE/ Insem. - 515

B.E. (Mechanical)

CAD/CAM & AUTOMATION

(2015 Course) (Semester - I) (402042)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right side indicates full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of scientific calculator allowed.

Q1) a) A Line PQ with P(1, 1), and Q(8, 8) is rotated through 20° in CW about point P. Find concatenated transformation matrix and new coordinates of line. Also, represent transformation graphically. **[6]**

b) Compare Geometric Transformation and Geometric Mapping. **[4]**

OR

Q2) A triangle PQR is represented by P(2,3), Q(10, 3) and R(10, 12). Find concatenated transformation matrix and new coordinates of triangle PQR, if

- a) Translation by 5 and 10 units resp. along X and Y axes.
- b) Scaled by 1.5 times along X axis and 0.5 times along Y axis.
- c) Rotated by 30° in CCW direction about perpendicular axis to XY plane and passing through origin O(0, 0).

Represent transformation graphically. **[10]**

Q3) a) What are the advantages of parametric representation of the curves in CAD system? **[4]**

b) The line L_1 has end points $P_1(2, 3, 4)$, $P_2(6, 8, 6)$ and line L_2 has end points $Q_1(1, 8, 4)$, $Q_2(8, 2, 6)$, then **[6]**

- i) Find the parametric equations of the lines.
- ii) Check, whether line L_1 is parallel to L_2 .

OR

P.T.O.

Q4) a) Write parametric equation of a circle with center $C(6, 4, 0)$ and Radius 6 units. Calculate coordinates of 3 equispaced points in first quadrant. [6]

b) Explain briefly parametric representation of analytical surfaces. [4]

Q5) The stepped bar shown in figure 1 is loaded axially by load $P = 50\text{kN}$, the modulus of elasticity of the bar is $0.7 \times 10^5 \text{ N/mm}^2$. Determine nodal displacement, elemental stress and reaction at support. [10]

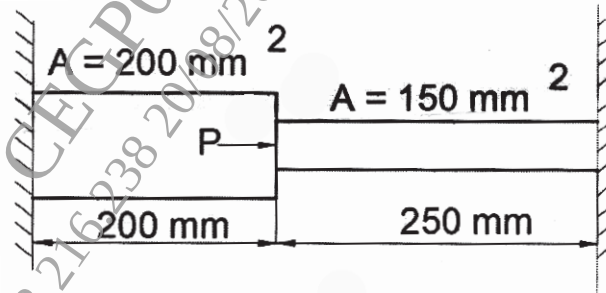


Figure 1: Qu 5

OR

Q6) a) For the Axially Loaded Spring System as shown in figure 2, determine [6]

- Nodal displacements,
- Deformation of each spring

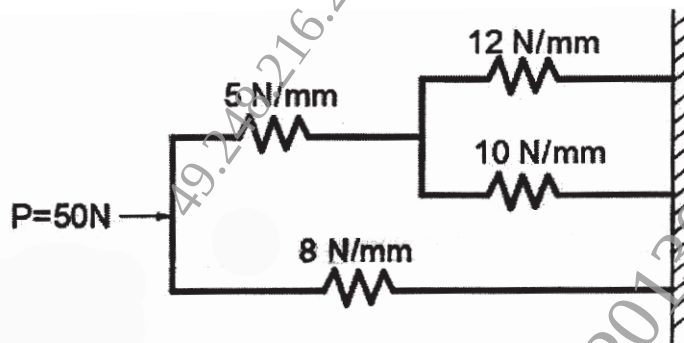


Figure 2: Qu 6(a)

b) Discuss Strain Displacement relations in FEA. [4]